Faculty of Engineering & Technology

Electrical and Computer Engineering Department

ENCS3130
LINUX LABORATORY
Project (1)

Medical Test Management System.

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- **Introduction**

This report shows the tests done on the Medical Test Management System, a shell script that manages patient test records. The system can handle several tasks, including adding new test results, updating existing records, deleting old entries, and searching for specific tests by patient ID, test type, or status. This report includes the test cases, input data, and results to confirm that the program works correctly.

- In the begin of the program the medical test file is contain this data:

Open medicaltest.txt ~/project

medicaltest.txt project.sh

```
Name: Hemoglobin (Hgb); Range: > 13.8, < 17.2; Unit: g/dL
Name: Blood Glucose Test (BGT); Range: > 70, < 99; Unit: mg/dL
Name: LDL Cholesterol Low-Density Lipoprotein (LDL); Range: < 100; Unit: mg/dL
Name: Systolic Blood Pressure (systole); Range: < 120; Unit: mm Hg
Name: Diastolic Blood Pressure (diastole); Range: < 80; Unit: mm Hg
Name: Blood Glucose Test (Bat); Range: > 70; Unit: mg/dL
```

- And the Recorded file is containing this date:

```
1300500: Hgb, 2024-03, 13.9, mg/dL, completed
1300511: LDL, 2024-03, 110, mg/dL, pending
1300500: LDL, 2024-03, 50, mg/dL, pending
1300511: Hgb, 2024-03, 14, mg/dL, completed
1300511: BGT, 2024-03, 80, mg/dL, Reviewed
1300500: Hgb, 2024-04, 20, mg/dL, completed
1300500: Bat, 2024-03, 80, mg/dL, Reviewed
1300500: LDL, 2024-7, 115, mg/dL, completed
```

- To start the process, we first run the shell script, which opens the following window:

```
ubuntu@ubuntu:~/project$ ./project.sh
1-Add a medical test:
2-search for a test by patient ID:
3-search for normal test:
4-Average test value:
5-Update a test result:
6-Delete a medical test:
7-Exist:

choose an operation
```

- The window shows options like adding a new test, searching for a test by patient ID, updating a test result, and more. From here, you can choose what you want to do.

1) After starting, I chose the first option to add a new medical test to our records:

1300501: diastole, 2024-05, 13.6, completed

As shown in the photo:

```
choose an operation
1
------------

Write ID:
1300501
Write test name:
diastole
Write the date (yyyy-mm):
2024-05
Write the test result (numeric or decimal):
75
Write the test status (pending, completed, Reviewed):
completed
------------

The test was added successfully.
------------

1-Add a medical test:
```

This is what the medical records look like after adding the new test:

```
1300500: Hgb, 2024-03, 13.9, mg/dL, completed
1300511: LDL, 2024-03, 110, mg/dL, pending
1300500: LDL, 2024-03, 50, mg/dL, pending
1300511: Hgb, 2024-03, 14, mg/dL, completed
1300511: BGT, 2024-03, 80, mg/dL, Reviewed
1300500: Hgb, 2024-04, 20, mg/dL, completed
1300500: Bat, 2024-03, 80, mg/dL, Reviewed
1300500: LDL, 2024-7, 115, mg/dL, completed
1300501: diastole, 2024-05, 75, mm Hg, completed
```

"The system determines if the test is accepted and available in our system (search if it in medical test) and ensures that it includes the correct unit of measurement from the medical test".

- Our system also checks if the input values are valid:
    1. Verifies if the ID is a number.
    2. Ensures the test is a string of characters.
    3. Confirms the result consists of digits.
    4. Checks if the date is in the correct format.
    5. Validates that the state is one of the three predefined states.

- Here I'm trying to invalid data:

```
choose an operation
1
____________

Write ID:
1300501a
Write test name:
BGT2
Write the date (yyyy-mm):
2024-06
Write the test result (numeric or decimal):
11a
Write the test status (pending, completed, Reviewed):
notcompleted
____________

Invalid ID. It should be numeric.
Write ID:
1300501
Invalid test name. It should contain only letters and spaces.
Write test name:
BGT
Invalid result. It should be numeric or a decimal number.
Write the test result (numeric or decimal):
11
Invalid status. It should be 'pending', 'completed', or 'Reviewed'.
Write the test status (pending, completed, Reviewed):
completed
The test was added successfully.
____________
```

- Here I'm trying to add a record, but the test does not exist in our system. (After verifying that the data is input correctly, the system will check if the test exists in our system)

```
choose an operation
1
____________

Write ID:
1300501
Write test name:
LYM
Write the date (yyyy-mm):
2024-05
Write the test result (numeric or decimal):
55
Write the test status (pending, completed, Reviewed):
completed
____________

Sorry, but our system does not support a test of this type: LYM.
____________

1-Add a medical test:
```

2) Search for a test by patient ID: the system will have the following functionality based on user selection:

1. Retrieve all patient tests.
2. Retrieve all up normal patient tests.
3. Retrieve all patient tests in a specific period.
4. Retrieve all patient tests based on test status.

2-1) Example of Retrieve all patient tests.

```
choose an operation
2
_____________

Write the patient ID:
1300500t
_____________

Invalid ID. It should be an integer.
reWrite the patient ID:
1300500
_____________

choose an operation
1-Retrieve all patient tests:
2-Retrieve all up normal patient tests:
3-Retrieve all patient tests in a given specific period:
4-Retrieve all patient tests based on test status:

1
_____________

1300500: Hgb, 2024-03, 13.9, mg/dL, completed
1300500: LDL, 2024-03, 50, mg/dL, pending
1300500: Hgb, 2024-04, 20, mg/dL, completed
1300500: Bat, 2024-03, 80, mg/dL, Reviewed
1300500: LDL, 2024-07, 115, mg/dL, completed
_____________
```

2-2) Example Retrieve all up normal patient tests.

```
choose an operation
2
____________

Write the patient ID:
1300500
____________

choose an operation
1-Retrieve all patient tests:
2-Retrieve all up normal patient tests:
3-Retrieve all patient tests in a given specific period:
4-Retrieve all patient tests based on test status:

2
____________

1300500: Hgb, 2024-03, 13.9, mg/dL, completed
1300500: LDL, 2024-03, 50, mg/dL, pending
1300500: Bat, 2024-03, 80, mg/dL, Reviewed
____________
```

2-3) Retrieve all patient tests in a specific period.

```
3
____________

Write the period yyyy-mm to yyyy-mm:
2024-04 to 2024-07
____________

1300500: Hgb, 2024-04, 20, mg/dL, completed
1300500: LDL, 2024-07, 115, mg/dL, completed
____________
```

2-4) Retrieve all patient tests based on test status.

```
choose an operation
1-Retrieve all patient tests:
2-Retrieve all up normal patient tests:
3-Retrieve all patient tests in a given specific period:
4-Retrieve all patient tests based on test status:

4
____________

Write the test status(pending,completed,Reviewed):
pending
____________

1300500: LDL, 2024-03, 50, mg/dL, pending
____________
```

"If the input is wrong it will ask user input again until give us one of three statuses"

```
choose an operation
1-Retrieve all patient tests:
2-Retrieve all up normal patient tests:
3-Retrieve all patient tests in a given specific period:
4-Retrieve all patient tests based on test status:

4
____________

Write the test status(pending,completed,Reviewed):
jj
____________

Invalid status. It should be 'pending', 'completed', or 'Reviewed'.
Write the test status (pending, completed, Reviewed):
completed
1300500: Hgb, 2024-03, 13.9, mg/dL, completed
1300500: Hgb, 2024-04, 20, mg/dL, completed
1300500: LDL, 2024-07, 115, mg/dL, completed
____________
```

3) Searching for up normal tests: the system will retrieve all up normal patients' tests based on the input medical test.

(It's like step 2 op 2, but applies to all tests, not just a specific one).

```
6-Delete a medical test:
7-Exist:

choose an operation
3
____________

1300500: Hgb, 2024-03, 13.9, mg/dL, completed
1300500: LDL, 2024-03, 50, mg/dL, pending
1300511: Hgb, 2024-03, 14, mg/dL, completed
1300511: BGT, 2024-03, 80, mg/dL, Reviewed
1300500: Bat, 2024-03, 80, mg/dL, Reviewed
1300501: diastole, 2024-05, 75, mm Hg, completed
____________

1-Add a medical test:
```

4) Average test value: the system will retrieve the average value of each medical test.

```
choose an operation
4
____________

Test  BGT average = 45.50
Test  Bat average = 80.00
Test  Hgb average = 15.96
Test  LDL average = 91.66
Test  diastole average = 75.00
____________
```

5) Update an existing test result.

```
choose an operation
5
_____________

Write the patient ID:
1300501
_____________

1300501: diastole, 2024-05, 75, mm Hg, completed
1300501: BGT, 2024-06, 11, mg/dL, completed

_____________

Write the test name:
BGT
_____________

1300501: BGT, 2024-06, 11, mg/dL, completed

_____________

enter the new value:
11.5
_____________

1300501: BGT, 2024-06, 11.5, mg/dL, completed
_____________
```

- In the end the system is display all Recorded

```
___________

enter the new value:
11.5
___________

1300501: BGT, 2024-06, 11.5, mg/dL, completed
___________

1300500: Hgb, 2024-03, 13.9, mg/dL, completed
1300511: LDL, 2024-03, 110, mg/dL, pending
1300500: LDL, 2024-03, 50, mg/dL, pending
1300511: Hgb, 2024-03, 14, mg/dL, completed
1300511: BGT, 2024-03, 80, mg/dL, Reviewed
1300500: Hgb, 2024-04, 20, mg/dL, completed
1300500: Bat, 2024-03, 80, mg/dL, Reviewed
1300500: LDL, 2024-07, 115, mg/dL, completed
1300501: diastole, 2024-05, 75, mm Hg, completed
1300501: BGT, 2024-06, 11.5, mg/dL, completed
___________
```

- If the input ID is not contained it return that is no input with this ID
- Example:

```
choose an operation
5
____________

Write the patient ID:
13005
____________

No tests found for the person with ID 13005.
____________
```

6) Delete a test

- First is display all recorded for you after is ask you to input the id for the patient.

```
choose an operation
6
____________

1300500: Hgb, 2024-03, 13.9, mg/dL, completed
1300511: LDL, 2024-03, 110, mg/dL, pending
1300500: LDL, 2024-03, 50, mg/dL, pending
1300511: Hgb, 2024-03, 14, mg/dL, completed
1300511: BGT, 2024-03, 80, mg/dL, Reviewed
1300500: Hgb, 2024-04, 20, mg/dL, completed
1300500: Bat, 2024-03, 80, mg/dL, Reviewed
1300500: LDL, 2024-07, 115, mg/dL, completed
1300501: diastole, 2024-05, 75, mm Hg, completed


____________

Write the patient ID:
1300500
____________

     1  1300500: Hgb, 2024-03, 13.9, mg/dL, completed
     2  1300500: LDL, 2024-03, 50, mg/dL, pending
     3  1300500: Hgb, 2024-04, 20, mg/dL, completed
     4  1300500: Bat, 2024-03, 80, mg/dL, Reviewed
     5  1300500: LDL, 2024-07, 115, mg/dL, completed
____________
```

- Then if there more than one line it asks user which line you want to delete and before it deletes it ask the user if they are sure they want to delete.

```
____________

     1  1300500: Hgb, 2024-03, 13.5, mg/dL, completed
     2  1300500: BGT, 2024-04, 60, mg/dL, Reviewed
     3  1300500: LDL, 2024-05, 30, mg/dL, pending
     4  1300500: Hgb, 2024-05, 40.5, mg/dL, Reviewed
     5  1300500: Hgb, 2024-03, 12, g/dL, Reviewed
____________

choose a line:
5
____________

Are you sure you want to delete(Y/N)?
Y
____________

1300500: Hgb, 2024-03, 13.5, mg/dL, completed
1300511: LDL, 2024-04, 110, mg/dL, pending
1300500: BGT, 2024-04, 60, mg/dL, Reviewed
1300511: BGT, 2024-04, 80, mg/dL, Reviewed
1300500: LDL, 2024-05, 30, mg/dL, pending
1300511: Hgb, 2024-05, 20, mg/dL, completed
1300500: Hgb, 2024-05, 40.5, mg/dL, Reviewed
1300512: Hgb, 2024-02, 13.9, g/dL, completed
1300512: LDL, 2024-09, 48, mg/dL, pending
1300400: Hgb, 2024-07, 56, g/dL, pending

delete file ended
____________
```

- If the user says no, it will exit the case.

Thank you.